A once-punctured torus contradicts the finite counting formula

Disproof of the explicit finite-count clause in problem 00000000568

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Claim and scope. The problem prints, in both languages, the necessary special case

$$N(g,1) = 6g - 3 + \sum_i 2^i \binom{2g}{i}.$$

Under the usual complete binomial sum, this gives $N(1,1) = 3 + (1 + 4 + 4) = 12$. We use the closed oriented torus with one interior marked point (one puncture). Its coefficient-free cluster algebra has infinitely many distinct cluster variables. The contradiction concerns this explicit finite formula, not the unspecified general Poincaré-coefficient formula and not a count of variables in a single cluster.

The published geometric model and its boundary. Fomin–Shapiro–Thurston, Example 4.6, gives the once-punctured torus matrix

$$B = \begin{pmatrix} 0 & 2 & -2 \\ -2 & 0 & 2 \\ 2 & -2 & 0 \end{pmatrix}.$$

One may obtain the standard ideal triangulation from a fundamental square with opposite sides identified and a diagonal: there are three arc labels and two oriented triangular faces, with the same cyclic order of labels on both faces. Each face contributes one cyclic signed adjacency matrix; their sum is B. The finite adjacency calculation is included in Lean. Definitions 5.1–5.2 specify seeds, mutations, all cluster variables and their generated algebra. Theorem 7.11, with its explicit closed-one-puncture exception, uses the *plain* component here; we do not combine the plain and notched components.

The identification of this exact published matrix model with the marked torus, and the geometric interpretation of its flips, are the cited external geometric definition bridge. Lean proves the complete algebraic statement for that exact model. It does not reprove the classification of surfaces or the arc-complex comparison theorem, and it introduces no axiom asserting those results.

The full seed model. Let $P = \mathbb{Q}[X_0, X_1, X_2]$ and $F = \mathrm{Frac}(P)$, using native multivariate polynomials and the native fraction field. The initial coordinates x_i are the actual images of X_i in F. They are algebraically independent and generate F as a field over $\mathbb{Q}$. A formal seed consists of its three coordinates, its exchange matrix B or $-B$, and proved algebraic independence and field generation. The native matrix mutation formula of Definition 4.7 is implemented, and every mutation sends B to $-B$ and $-B$ to B. For any index k (indices modulo 3), its coordinate mutation is

$$u_k' = \frac{u_{k+1}^2 + u_{k+2}^2}{u_k}, \qquad u_i' = u_i \quad (i \neq k).$$

This is proved equal to the usual positive-column monomial plus negative-column monomial exchange formula for the actual exchange matrix. All denominators are nonzero by algebraic independence. The numerator is nonzero because it is the evaluation of a nonzero polynomial; evaluating that polynomial at $(1,1,1)$ gives 2.

Every mutation preserves field generation: the unchanged coordinates and the new coordinate recover the old coordinate by the same exchange equation. A generating triple in F, whose transcendence degree is at least 3 by the initial independent triple, is algebraically independent. This is proved using native transcendence basis results. Mutations are actual involutions, not relations imposed by an axiom. Consequently every finite word in *all three* mutation directions gives a valid seed. Let $\mathcal{X} \subset F$ be precisely the set of coordinates of all these seeds. The algebra is the actual native $\mathbb{Z}$-subalgebra $\mathbb{Z}[\mathcal{X}] \subset F$. Thus the coefficient-free convention is used with integer ground ring; the variable set has not been rationalized or replaced by word labels. Coefficients in the exchange relation are both 1, as for the trivial coefficient semifield.

A legal specialization and its invariant. Use the native localization

$$L = P[(X_0X_1)^{-1}].$$

It injects into F. There is also a genuine ring homomorphism $e : L \to \mathbb{Q}$ sending each X_i to 1, because the sole inverted generator X_0X_1 maps to the unit 1. No field homomorphism $F \to \mathbb{Q}$ with these values is used or claimed. Write x, y, z for the local coordinate images and set

$$c = (x^2 + y^2 + z^2)(xy)^{-1}, \qquad f_0 = x, \quad f_1 = y, \quad f_{n+2} = cf_{n+1} - f_n \quad \text{in } L.$$

The localization inverse gives $cxy = x^2 + y^2 + z^2$. Induction proves the genuine polynomial invariant and exchange identity

$$f_n^2 + f_{n+1}^2 + z^2 = cf_nf_{n+1}, \qquad f_nf_{n+2} = f_{n+1}^2 + z^2.$$

For $a_n = e(f_n)$, these definitions imply $a_0 = a_1 = 1$ and $a_{n+2} = 3a_{n+1} - a_n$. Induction gives $0 < a_{n+1}$ and $a_n \le a_{n+1}$ for all n. Hence $a_{n+2} \ge 2a_{n+1} > a_{n+1}$, and $n \mapsto a_{n+1}$ is strictly increasing. In particular all f_n and their images in F are nonzero.

The infinite family consists of actual cluster variables. Let $\bar{f}_n$ denote the image of f_n in F. Starting with the initial seed, repeat mutation in direction 0 and then direction 1, a total of r pairs. The proved exchange identities and nonzero denominators give its exact cluster

$$(\bar{f}_{2r},\ \bar{f}_{2r+1},\ x_2).$$

This path is embedded in the full finite-word seed model; direction 2 is still available in that model. Thus $\bar{f}_{2r+1} \in \mathcal{X}$ for every r. If two of these actual rational functions were equal, injectivity of $L \to F$ would give equality in L. Applying e would give $a_{2r+1} = a_{2s+1}$, so strict increase yields $r = s$. This is an injection $\mathbb{N} \hookrightarrow \mathcal{X}$ and proves native `Set.Infinite`. In fact the proof excludes every finite set representing $\mathcal{X}$, for every proposed finite cardinal N, and therefore excludes 12. It never uses the default value of `Nat.card` on an infinite type.

Formal verification and limits. The complete proof ends with the following theorem:

`lean/Main.lean:  complete_counterexample`

Use Lean 4.33.0 and the pinned Mathlib revision

`db584cd6d46c92f209a44c0f1c829460d327499d`.

From the `lean/` directory run the following commands, or run `lean -version` followed by `./verify.sh` for the last three checks:

```
lean --version
lake build +Main
lake env lean -DwarningAsError=true Check.lean
lake env leanchecker --fresh Main
```

All actual namespace declarations, including structure-generated auxiliaries, are enumerated. Only `propext`, `Classical.choice` and `Quot.sound` are allowed. The finite adjacency checks use ordinary kernel decision proofs. There is no `sorry`, native decision evaluation, or extra axiom. Exact commands, source hashes, raw logs and execution observations accompany the submission. Production timings are diagnostic and are not controlled research-efficiency evidence. The full original bilingual source is retained.

References

[1] S. Fomin, M. Shapiro and D. Thurston, *Cluster algebras and triangulated surfaces. Part I: Cluster complexes*, Acta Mathematica **201** (2008), 83–146. arXiv:math/0608367v3, printed pp.17–18 (Example 4.6, Definitions 4.7 and 5.1–5.2) and p.28 (Proposition 7.10 and Theorem 7.11). These selected pages were read; no whole-paper read is claimed.